AI Study Assistant – RAG Demo Notes

Introduction of project (AI Study Assistant):

An AI Study Assistant is a software system that helps users understand large documents by answering questions. Instead of manually searching through pages, users can ask questions and get answers directly from the document.

This project is based on **Retrieval-Augmented Generation**, commonly known as **RAG**. RAG combines information retrieval with question answering to produce accurate and grounded answers.

What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation is an approach where a system **first retrieves relevant information from documents** and **then uses that retrieved information to answer a question**.

In a RAG system:

- The answer is based only on the document content
- The system does not generate random or unrelated information

In this project, RAG is used to answer questions from lecture notes and study material.

Why RAG is Needed

Large documents contain a lot of information. Reading the entire document to find one answer is time-consuming.

RAG helps by:

- Finding only the relevant parts of the document
- Answering questions using those relevant parts
- Reducing the time needed to study or revise

Key Components of the RAG Pipeline

The RAG pipeline consists of several important steps. Each step plays a role in answering questions correctly.

Steps involved in RAG pipeline:

1.Document Upload

2.Text Extraction

3.Text Chunking

4.TF-IDF Vectorization

5.FAISS Retrieval

6.Answer Generation

Step 1: Document Upload

In the first step, the user uploads a PDF document. This document can be lecture notes, study material, or technical documentation.

The uploaded document becomes the knowledge source for answering questions.

Step 2: Text Extraction

After uploading the PDF, the system extracts text from the document. Text extraction converts the PDF content into readable text that can be processed by the system.

This step is important because machine learning models work on text, not PDFs.

Step 3: Text Chunking

The extracted text is split into smaller parts called **chunks**. Each chunk contains a limited number of words.

Chunking is done because:

- Large documents cannot be processed as a single block
- Smaller chunks preserve context
- Chunking improves retrieval accuracy

Step 4: TF-IDF Vectorization

TF-IDF stands for **Term Frequency – Inverse Document Frequency**. It is a technique used to convert text into numerical vectors.

TF-IDF assigns higher importance to words that:

- Appear frequently in a chunk
- Are important for understanding the content

In this project, TF-IDF is used to represent document chunks and questions numerically.

Step 5: FAISS Retrieval

FAISS is a library used for fast similarity search. It helps retrieve the most relevant chunks based on the user’s question.

When a question is asked:

- FAISS compares the question vector with chunk vectors
- The most similar chunks are retrieved

FAISS makes the retrieval process fast and efficient.

Step 6: Answer Generation

After retrieving the relevant chunks, the system selects the most relevant sentence or paragraph. This sentence becomes the final answer.

This approach is called **extractive question answering** because:

- The answer is extracted from the document
- No new information is generated

Advantages of This RAG-Based Approach

This RAG-based AI Study Assistant has several advantages.

First, the answers are **grounded in the document**. This means the system does not make up information.

Second, the approach is **transparent and explainable**. The user can see the source context from which the answer was taken.

Third, the system reduces study time by providing direct answers instead of manual searching.

Type of RAG Used in This Project

This project uses **extractive RAG**. In extractive RAG:

- The system retrieves information
- The system selects answers directly from the document

This is different from generative RAG, where large language models generate new text.

Conclusion

This AI Study Assistant demonstrates how Retrieval-Augmented Generation can be used to build intelligent question-answering systems. By combining text retrieval and extractive answering, the system helps users interact efficiently with large educational documents.